

GD - HW2

THEORETICAL PROBLEMS

1 Choose the simplest function:

$$f(x) = \frac{1}{2} L x^2, L > 0$$

$$\nabla f(x) = Lx$$

Linear recurrences

$$p\text{-periodic: } x_{k+p} = x_k \Rightarrow r^p = 1$$

1) 2-periodic

$$r^2 = 1 \rightarrow r = 1 \text{ (} -1 \text{)} \text{ since we want non-constant}$$

2) 3-periodic

$$r^3 = 1$$

$$r^3 - 1 = 0$$

$$(r-1)(r^2+r+1) = 0$$

$r=1$ gives constant seq.

$$\Rightarrow r^2 + r + 1 = 0$$

$$r = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot 1}}{2} = -\frac{1}{2} \pm i \frac{\sqrt{3}}{2}$$

$$r = e^{\pm i 2\pi/3}$$

A) GD

$$\text{recurrence: } x_{k+1} = x_k - \gamma L x_k = (1 - \gamma L) x_k$$

$$2\text{-periodic: need } x_{k+1} = -x_k \Rightarrow (1 - \gamma L) = -1 \Rightarrow \gamma L = 2 \rightarrow$$

☒ $\rightarrow$ since $\gamma, L > 0$ this is valid.

$$1, -1, 1, -1, \dots$$

3-periodic: For 1-step recurrence to be 3-periodic,
 $\boxed{X}$ the multiplier $(1-\gamma L)$ should be $e^{i2\pi/3}$

$$\rightarrow (1-\gamma L) \in \mathbb{R}$$

$\rightarrow$ only real 3rd root of unity is 1 $\Rightarrow \gamma L = 0$
 $\Downarrow$

GID cannot $\Leftarrow$ constant
cycle in threes

B) Polya's GID

$$x_{k+1} = x_k - \gamma L x_k + \mu (x_k - x_{k-1})$$

$$x_{k+1} - (1 - \gamma L + \mu) x_k + \mu x_{k-1} = 0$$

$$r_1, r_2 \text{ satisfy: } \underline{r^2 - (1 - \gamma L + \mu)r + \mu = 0}$$

2-periodic: need $r_1, r_2 = -1$

$\swarrow$ product of roots
= μ

$$\boxed{\checkmark} \quad \mu = (-1)(-1) = 1$$

$$\text{sum: } 1 - \gamma L + 1 = -2$$

$$\Rightarrow \gamma L = 4$$

$$\text{sum of roots} \\ = 1 - \gamma L + \mu$$

$\mu, \gamma > 0$, this works

3-periodic: need complex roots $e^{\pm i2\pi/3}$

$$\boxed{\checkmark} \quad \mu = 1$$

$$\text{sum: } 1 - \gamma L + \mu = -1 \Rightarrow 1 - \gamma L + 1 = -1$$

$$\gamma L = 3$$

$\rightarrow$ Polya can "rotate". $\mu=1, \gamma L=3$ makes it cycle
rotate in triangle

c) Nesterov GD

$$x_{k+1} = x_k - \gamma L x_k = (1 - \gamma L) x_k$$

$$x_{k+1} = (1 - \gamma L) (x_k + \mu (x_k - x_{k-1}))$$

$$x_{k+1} - (1 - \gamma L)(1 + \mu)x_k + \mu(1 - \gamma L)x_{k-1} = 0$$

product of roots = $\mu(1 - \gamma L)$

sum of roots = $(1 - \gamma L)(1 + \mu)$

2-periodic: $x_k = (-1)^k$

✓ $-1 - (1 - \gamma L)(1 + \mu)(1) + \mu(1 - \gamma L)(-1) = 0$

$$-1 - (1 - \gamma L)(1 + 2\mu) = 0 \Rightarrow 1 - \gamma L = \frac{-1}{1 + 2\mu}$$

positive

$1 - \gamma L$ is a small neg. number

$$\Rightarrow \underline{\gamma L > 1} \text{ - valid } \checkmark$$

3-periodic: need $e^{\pm i 2\pi/3}$

product: $\mu(1 - \gamma L) = 1$

sum: $(1 - \gamma L)(1 + \mu) = -1$

✗ $\left. \begin{array}{l} \text{product: } \mu(1 - \gamma L) = 1 \\ \text{sum: } (1 - \gamma L)(1 + \mu) = -1 \end{array} \right\} \frac{(1 - \gamma L)(1 + \mu)}{\mu(1 - \gamma L)} = \frac{-1}{1}$

$$\frac{1 + \mu}{\mu} = -1$$

$$1 + \mu = -\mu$$

$$\underline{\mu = -0.5} \Rightarrow \text{not pos.}$$

GD: May result in 2-periodic seq.

Polyak GD: May result in 2- and 3-periodic seq.

Nesterov GD: May result in 2-periodic seq

2.1 determine optimal γ, μ for Polyak GD for:

$$f(x, y, z) = x^2 + 2y^2 - 2yz + 4z^2 + 3x - 4y + 5z$$

Theorem: for ~~quad~~ quad. func., P. GD with params.

$$\mu = \frac{\sqrt{\beta} - \sqrt{\alpha}}{\sqrt{\beta} + \sqrt{\alpha}}, \quad \gamma = \frac{4}{(\sqrt{\alpha} + \sqrt{\beta})^2} \quad \text{satisfies:} \quad \left| \begin{array}{l} \alpha = \min \lambda \\ \beta = \max \lambda \\ \text{of } \nabla^2 f \end{array} \right.$$
$$\|x_{k+1} - x^*\| \leq \left(\frac{\sqrt{\beta} - \sqrt{\alpha}}{\sqrt{\beta} + \sqrt{\alpha}} + \varepsilon^k \right)^k \|x_1 - x^*\|$$

Hessian:

$$\begin{array}{l} \frac{\partial f}{\partial x} = 2x + 3 \\ \frac{\partial f}{\partial y} = 4y - 4 - 2z \\ \frac{\partial f}{\partial z} = -2y + 8z + 5 \end{array} \quad \left\{ \begin{array}{lll} \frac{\partial^2 f}{\partial x^2} = 2 & \frac{\partial^2 f}{\partial y \partial x} = 0 & \frac{\partial^2 f}{\partial z \partial x} = 0 \\ \frac{\partial^2 f}{\partial x \partial y} = 0 & \frac{\partial^2 f}{\partial y^2} = 4 & \frac{\partial^2 f}{\partial z \partial y} = -2 \\ \frac{\partial^2 f}{\partial x \partial z} = 0 & \frac{\partial^2 f}{\partial y \partial z} = -2 & \frac{\partial^2 f}{\partial z^2} = 8 \end{array} \right.$$

$$H = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & -2 \\ 0 & -2 & 8 \end{bmatrix}$$

$$\det(H - \lambda I) = 0$$

$$\rightarrow \begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & 4-\lambda & -2 \\ 0 & -2 & 8-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)((4-\lambda)(8-\lambda) - 4) = 0$$

$$(2-\lambda)(32 - 12\lambda + \lambda^2 - 4) = 0$$

$$(2-\lambda)(\lambda^2 - 12\lambda + 28) = 0$$

$$\downarrow$$

$$\underline{\lambda_1 = 2}$$

$$\hookrightarrow \lambda_{2,3} = \frac{12 \pm \sqrt{144 - 4 \cdot 1 \cdot 28}}{2} = \frac{12 \pm \sqrt{32}}{2}$$

$$= \frac{12 \pm 4\sqrt{2}}{2} = 6 \pm 2\sqrt{2}$$

(function is strictly convex)

$$\underline{\lambda_2 = 6 - 2\sqrt{2} \approx 3,17}$$

$$\underline{\lambda_3 = 6 + 2\sqrt{2} \approx 8,83}$$

$$\Rightarrow \underline{\alpha = \lambda_1 = 2}, \quad \underline{\beta = \lambda_3 = 6 + 2\sqrt{2}}$$

optimal μ : ~~μ~~ $\mu = \left(\frac{\sqrt{6+2\sqrt{2}} - \sqrt{2}}{\sqrt{6+2\sqrt{2}} + \sqrt{2}} \right)^2 \approx \underline{\underline{0,126}}$

optimal γ : $\gamma = \frac{4}{(\sqrt{2} + \sqrt{6+2\sqrt{2}})^2} \approx \underline{\underline{0,208}}$